

Hence, graph should be quadratic.

9 Ans: A

$$T \cos 40^\circ = mg \quad \text{----- (1)}$$

$$T \sin 40^\circ = mv^2 / r \quad \text{----- (2)}$$

$$(2) / (1), \quad \tan 40^\circ = v^2 / rg$$

$$v^2 = rg \tan 40^\circ = (1.2 \sin 40^\circ)(9.81)(\tan 40^\circ)$$

$$v = 2.5 \text{ m s}^{-1}$$

10 Ans: D